Tao Chen

Optical thermal metrology · micro/nanoscale heat transport

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Education

M.Eng. in Energy and Power Engineering 2023 – 2026

Huazhong University of Science and Technology | Advisor: Prof. Puqing Jiang

- Admitted ranked 1st in the entrance examination, with a perfect score in mathematics.
- Thesis: Development and application of a thermal-property measurement technique based on the square-pulsed source (SPS) method.
- Selected coursework: Theory and Engineering Applications of Heat Transfer; Advanced Applied Engineering Mathematics; Thermal Signal Processing and Visualization; Modern Instrumental Analysis Methods; Academic Writing.

B.Eng. in Energy and Power Engineering 2019 – 2023

Huazhong University of Science and Technology

- Thesis: Development and application of an infrared-based AC calorimetry system.
- Selected coursework: Engineering Heat Transfer; Engineering Thermodynamics; Engineering Numerical Methods; Calculus; Linear Algebra; Probability and Statistics.

Research Interests

- Optical thermal metrology (TDTR, FDTR, SDTR, and the square-pulsed source method) for micro/nanoscale heat transfer and non-equilibrium thermal transport.
- Thermal characterization of thin films, interfaces, anisotropic and two-dimensional materials, liquids, and thermal interface materials.
- Inverse problems, uncertainty quantification, and the interplay between heat transfer and applied mathematics.

Research Experience

Optical Measurement of Multilayer-Film Thermal Properties in Chips 2025 – Present

Huazhong University of Science and Technology (with an industry partner)

- Measure the thermal conductivity and volumetric heat capacity of each micron-thick layer in multilayer chip stacks.
- Built the optical platform and an automated routine for spatial thermal-property mapping.

Near-Junction Thermal Transport in GaN Devices 2024 – Present

Huazhong University of Science and Technology

- Developed the square-pulsed source (SPS) thermoreflectance method (see Publications).
- Extending it to an ultraviolet square-pulse transient scheme for *in situ*, near-junction measurements.
- Quantifying how growth conditions set the thermal resistance of passivation layers and interfaces.

Micro/Nanoscale Thermal Characterization for 3D-Stacked Chips 2024

Huazhong University of Science and Technology (with an industry partner)

- Designed and built an optical micro/nanoscale thermal-characterization platform with <10% measurement error and automated spatial mapping.
- Linked micro/nanoscale structural features to thermal transport across real chip samples to inform 3D-stacked chip design.

Honors and Awards

National Graduate Scholarship, Ministry of Education of China (national)	Dec 2025
First-Class Postgraduate Scholarship, Huazhong University of Science and Technology	Dec 2025
National Graduate Scholarship, Ministry of Education of China (national)	Dec 2024
First-Class Postgraduate Scholarship, Huazhong University of Science and Technology	Dec 2024
Merit Student, Huazhong University of Science and Technology	Dec 2024
Outstanding Conference Paper Award, 3rd Chinese Conference on Thermal Properties	Jun 2024
Self-Reliance Scholarship, School of Energy and Power Engineering	Feb 2020

Professional Service

Referee for *Physical Review Letters*, *Physical Review Applied*, and *Physical Review Materials*.

Journal Articles

Chen, T., Xiao, B., Qian, X., & Jiang, P. (2026). Simultaneous measurement of pressure-dependent bulk and interfacial thermal properties in thermal interface materials using square-pulsed source thermoreflectance. *Applied Thermal Engineering* (under review). arXiv:2603.22733.

Chen, T., Wang, D., & Jiang, P. (2026). Physics-derived natural coordinates for fast uncertainty propagation in multilayer thermoreflectance. *International Journal of Heat and Mass Transfer* (under review).

Chen, T., & Jiang, P. (2025). Weak coupling of diffusional and phonon-like modes in liquids revealed by dynamic Kapitza length. *Physical Review Letters* (under revision).

Shen, Y., **Chen, T.**, Song, S., Zhang, K., Chen, Y., Zhao, L., & Jiang, P. (2026). Mechanistic insight into BEOL thermal transport via optical metrology and multiphysics simulation. *International Communications in Heat and Mass Transfer*, 178, 111751.

Chen, T., & Jiang, P. (2025). Universal thermometry of solid–liquid interfacial thermal conductance. *Applied Physics Letters*, 127, 191608. [Featured Article]

Bao, Y., **Chen, T.**, Miao, Z., Zheng, W., Jiang, P., Chen, K., Guo, R., & Xue, D. (2025). Machine-learning-assisted understanding of depth-dependent thermal conductivity in lithium niobate induced by point defects. *Advanced Electronic Materials*, 11(11), 2400944.

Xiao, B., **Chen, T.**, Zhang, W., Qian, X., & Jiang, P. (2025). Fast and reliable thermal property extraction from FDTR measurements via hybrid particle swarm optimization. *International Journal of Heat and Mass Transfer*, 256, 128109.

Chen, T., & Jiang, P. (2025). Simultaneous, non-contact measurement of liquid and interfacial thermal properties via a differential square-pulsed source method. *Langmuir*, 41(38), 26496–26504.

Zhang, K., **Chen, T.**, Ma, J., & Jiang, P. (2025). A variable-spot-size and multi-frequency square-pulsed source (SPS) approach for comprehensive characterization of anisotropic thermal transport properties in multilayered thin films. *Journal of Physics D: Applied Physics*, 58(32), 325302.

Zhang, M., **Chen, T.**, Song, S., Bao, Y., Guo, R., Zheng, W., Jiang, P., & Yang, R. (2025). Extending the low-frequency limit of time-domain thermoreflectance via periodic waveform analysis. *Journal of Applied Physics*, 138(5), 055101.

Chen, T., & Jiang, P. (2025). A measurement method for the thermal conductivity and specific heat of submillimeter-sized low-thermal-conductivity materials. *Journal of Huazhong University of Science and Technology (Natural Science Edition)*, 53(7), 88–93.

Chen, T., & Jiang, P. (2025). Decoupling thermal properties in multilayered systems for advanced thermoreflectance experiments. *Physical Review Applied*, 23(4), 044004.

Zhang, M., **Chen, T.**, Zeng, A., Tang, J., Guo, R., & Jiang, P. (2025). Simultaneous measurement of thermal conductivity, heat capacity, and interfacial thermal conductance by leveraging negative delay-time data in time-domain thermoreflectance. *Progress in Natural Science: Materials International*, 35(2), 375–384.

Song, S., **Chen, T.**, & Jiang, P. (2025). Comprehensive thermal property measurement of semiconductor heterostructures using the square-pulsed source (SPS) method. *Journal of Applied Physics*, 137(5), 055101.

Chen, T., & Jiang, P. (2025). An optical method for measuring local convective heat transfer coefficients over $100\,\mathrm{W/(m^2{\cdot}K)}$ with sub-millimeter resolution. *Measurement*, 249, 117001.

Chen, T., & Jiang, P. (2024). Unraveling the intrinsic relationship of thermal properties in thermoreflectance experiments. *Acta Physica Sinica*, 73(23), 230202. [Cover Article]

Chen, T., Song, S., Hu, R., & Jiang, P. (2024). Comprehensive measurement of the three-dimensional thermal conductivity tensor using a beam-offset square-pulsed source (BO-SPS) approach. *International Journal of Thermal Sciences*, 207, 109347.

Chen, T., Song, S., Shen, Y., Zhang, K., & Jiang, P. (2024). Simultaneous measurement of thermal conductivity and heat capacity across diverse materials using the square-pulsed source (SPS) technique. *International Communications in Heat and Mass Transfer*, 158, 107849.

Conference Presentations

Chen, T., & Jiang, P. (2025, Oct.). Pressure-dependent thermal properties of thermal interface materials [oral presentation]. 9th Workshop on Thermal Transport (WTT 2025), China.

Chen, T., Hu, Z., & Jiang, P. (2024, Dec.). Measurement of the three-dimensional anisotropic thermal conductivity tensor of micro-sized samples using the beam-offset square-pulsed source (BO-SPS) method; and Accurate measurement of the local intrinsic convective heat transfer coefficient using the square-pulsed source (SPS) method [two oral presentations]. Annual Conference on Heat and Mass Transfer, Chinese Society of Engineering Thermophysics, China.

Chen, T., & Jiang, P. (2024, Sep.). Revealing the intrinsic relationship of thermal-property parameters in thermoreflectance experiments [oral presentation]. 3rd Chinese Conference on Thermal Properties, China. *(Outstanding Conference Paper Award)*

Chen, T., Song, S., & Jiang, P. (2024, Jun.). Advancing thermal characterization: simultaneous micrometer-scale measurement of thermal conductivity and heat capacity in low-conductivity materials using the square-pulsed source (SPS) technique [oral presentation]. 3rd Asian Conference on Thermal Sciences (ACTS 2024), Shanghai, China.

Patents

Jiang, P., & **Chen, T.** (2025). Method and system for measuring the local convective heat transfer coefficient. Patent No. ZL 2024 1 0763998.4.

Jiang, P., **Chen, T.**, & Xiao, B. (2025). System and method for measuring the thermal properties of soft-matter thin films. Patent Application No. 2025109077810.

Jiang, P., Yang, R., Song, S., **Chen, T.**, & Wu, J. (2024). Method and system for measuring the anisotropic thermal properties of sub-millimeter-scale samples. Patent No. ZL 2024 1 0057927.2.

Industry Experience

Engineering Intern, Sichuan Air Separation Equipment (Group) Co., Ltd., Chengdu, China 2022

- Studied the design and manufacture of cryogenic equipment (pressure vessels, compressors, heat exchangers) and the thermodynamics of air-separation units.
- Authored a technical report summarizing key engineering findings.

Skills

Experimental: Design, build, and calibrate optical thermal-metrology systems; TDTR, FDTR, SDTR, and SPS measurements.

Computational: MATLAB, Python, COMSOL, Origin, LaTeX, SolidWorks; modeling, inverse fitting, and uncertainty analysis.

Languages: English (proficient), Mandarin (native).